A REPRINT SERIES IN PUBLIC HEALTH HISTORY

REPRINT NO. 1

Surveillance Notes

Two reports of June 5 and July 3, 1981: the first published accounts of what became known as AIDS, as the Centers for Disease Control printed them

1 *Pneumocystis* Pneumonia — Los Angeles
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EPIDEMIOLOGIC NOTES AND REPORTS

Pneumocystis Pneumonia — Los Angeles

June 5, 1981 · Vol. 30, No. 21 · pp. 250–252

In the period October 1980-May 1981, 5 young men, all active homosexuals, were treated for biopsy-confirmed *Pneumocystis carinii* pneumonia at 3 different hospitals in Los Angeles, California. Two of the patients died. All 5 patients had laboratory-confirmed previous or current cytomegalovirus (CMV) infection and candidal mucosal infection. Case reports of these patients follow.

Figure 1 and Table 1 were added for this reprint. They restate the dates and findings of the five case reports below and add nothing to them (Figure 1; Table 1).

Patient 1: A previously healthy 33-year-old man developed *P. carinii* pneumonia and oral mucosal candidiasis in March 1981 after a 2-month history of fever associated with elevated liver enzymes, leukopenia, and CMV viruria. The serum complement-fixation CMV titer in October 1980 was 256; in May 1981 it was 32.* The patient's condition deteriorated despite courses of treatment with trimethoprim-sulfamethoxazole (TMP/SMX), pentamidine, and acyclovir. He died May 3, and postmortem examination showed residual *P. carinii* and CMV pneumonia, but no evidence of neoplasia.

Patient 2: A previously healthy 30-year-old man developed *P. carinii* pneumonia in April 1981 after a 5-month history of fever each day and of elevated liver-function tests, CMV viruria, and documented seroconversion to CMV, i.e., an acute-phase titer of 16 and a convalescent-phase titer of 28* in anticomplement immunofluorescence tests. Other features of his illness included leukopenia and mucosal candidiasis. His pneumonia responded to a course of intravenous TMP/SMX, but, as of the latest reports, he continues to have a fever each day.

Patient 3: A 30-year-old man was well until January 1981 when he developed esophageal and oral candidiasis that responded to Amphotericin B treatment. He was hospitalized in February 1981 for *P. carinii* pneumonia

* Paired specimens not run in parallel.

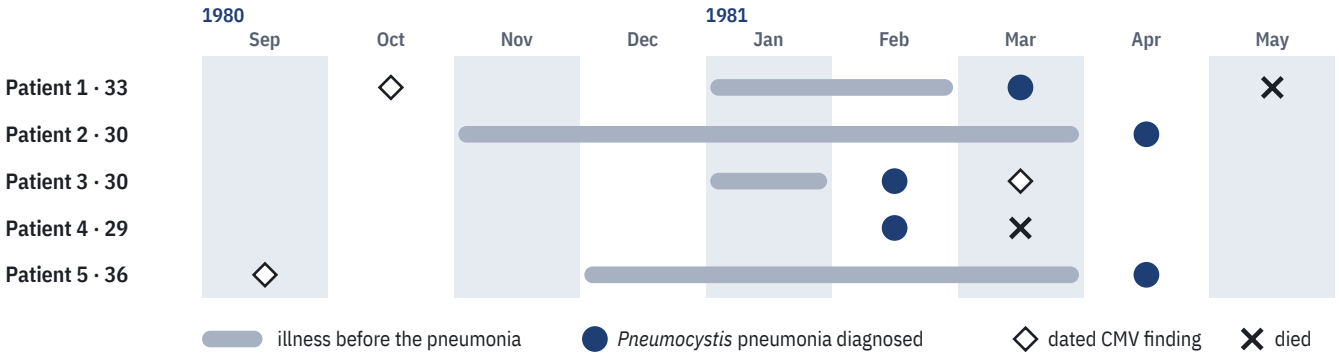


Figure 1. The five patients, September 1980 to May 1981, by month. The illness before the pneumonia is counted back from the month of diagnosis, as each report gives it.

Drawn by the editors. Compiled for this reprint from the case reports; not part of the 1981 report.

Table 1. The five case reports at a glance.

Patient	Age	Pneumonia	CMV evidence	Other findings	Course
1	33	Mar 1981	CF titer 256 (Oct 1980), 32 (May 1981); viruria	Oral candidiasis; leukopenia; raised liver enzymes	Died May 3
2	30	Apr 1981	Seroconversion (16 to 28); viruria	Mucosal candidiasis; leukopenia	Pneumonia responded; daily fever
3	30	Feb 1981	CF titer 8 (Mar 1981); esophageal biopsy positive	Esophageal and oral candidiasis	Pneumonia responded
4	29	Feb 1981	CMV in lung tissue at postmortem	Hodgkins disease 3 years earlier, treated	Died in March
5	36	Apr 1981	Clinical CMV infection (Sep 1980); retinitis; CF titer 128	Oral candidiasis; sulfa-induced neutropenia	Under treatment

Compiled for this reprint from the case reports; not part of the 1981 report.

that responded to oral TMP/SMX. His esophageal candidiasis recurred after the pneumonia was diagnosed, and he was again given Amphotericin B. The CMV complement-fixation titer in March 1981 was 8. Material from an esophageal biopsy was positive for CMV.

Patient 4: A 29-year-old man developed *P. carinii* pneumonia in February 1981. He had had Hodgkins disease 3 years earlier, but had been successfully treated with radiation therapy alone. He did not improve after being given intravenous TMP/SMX and corticosteroids and died in March. Postmortem examination showed no evidence of Hodgkins disease, but *P. carinii* and CMV were found in lung tissue.

Patient 5: A previously healthy 36-year-old man with a clinically diagnosed CMV infection in September 1980 was seen in April 1981 because of a 4-month history of fever, dyspnea, and cough. On admission he was found to have *P. carinii* pneumonia, oral candidiasis, and CMV retinitis. A complement-fixation CMV titer in April 1981 was 128. The patient has been treated with 2 short courses of TMP/SMX that have been limited because of a sulfa-induced neutropenia. He is being treated for candidiasis with topical nystatin.

The diagnosis of *Pneumocystis* pneumonia was confirmed for all 5 patients antemortem by closed or open lung biopsy. The patients did not know each other and had no known common contacts or knowledge of sexual partners who had had similar illnesses. The 5 did not have comparable histories of sexually transmitted disease. Four had serologic evidence of past hepatitis B infection but had no evidence of current hepatitis B surface antigen. Two of the 5 reported having frequent homosexual contacts with various partners. All 5 reported using inhalant drugs, and 1 reported parenteral drug abuse. Three patients had profoundly depressed numbers of thymus-dependent lymphocyte cells and profoundly depressed *in vitro* proliferative responses to mitogens and antigens. Lymphocyte studies were not performed on the other 2 patients.

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Editorial Note: *Pneumocystis* pneumonia in the United States is almost exclusively limited to severely immunosuppressed patients (1). The occurrence of pneumocystosis in these 5 previously healthy individuals without a clinically apparent underlying immunodeficiency is unusual. The fact that these patients were all homosexuals suggests an association between some aspect of a homosexual lifestyle or disease acquired through sexual contact and *Pneumocystis* pneumonia in this population. All 5 patients described in this report had laboratory-confirmed CMV disease or virus shedding within 5 months of the diagnosis of *Pneumocystis* pneumonia. CMV infection has been shown to induce transient abnormalities of *in vitro* cellular-immune function in otherwise healthy human hosts (2, 3). Although all 3 patients tested had abnormal cellular-immune function, no definitive conclusion regarding the role of CMV infection in these 5 cases can be reached because of the lack of published data on cellular-immune function in healthy homosexual males with and without CMV antibody. In 1 report, 7 (3.6%) of 194 patients with pneumocystosis also had CMV infection; 40 (21%) of the same group had at least 1 other major concurrent infection (1). A high prevalence of CMV infections among homosexual males was recently reported: 179 (94%) of 190 males reported to be exclusively homosexual had serum antibody to CMV, and 14 (7.4%) had CMV viruria; rates for 101 controls of similar age who were reported to be exclusively heterosexual were 54% for

seropositivity and zero for viruria (4). In another study of 64 males, 4 (6.3%) had positive tests for CMV in semen, but none had CMV recovered from urine. Two of the 4 reported recent homosexual contacts. These findings suggest not only that virus shedding may be more readily detected in seminal fluid than in urine, but also that seminal fluid may be an important vehicle of CMV transmission (5).

All the above observations suggest the possibility of a cellular-immune dysfunction related to a common exposure that predisposes individuals to opportunistic infections such as pneumocystosis and candidiasis. Although the role of CMV infection in the pathogenesis of pneumocystosis remains unknown, the possibility of *P. carinii* infection must be carefully considered in a differential diagnosis for previously healthy homosexual males with dyspnea and pneumonia.

REFERENCES

1. Walzer PD, Perl DP, Krogstad DJ, Rawson PG, Schultz MG. *Pneumocystis carinii* pneumonia in the United States. Epidemiologic, diagnostic, and clinical features. *Ann Intern Med*. 1974;80:83-93.
2. Rinaldo CR Jr, Black PH, Hirsch MS. Interaction of cytomegalovirus with leukocytes from patients with mononucleosis due to cytomegalovirus. *J Infect Dis*. 1977;136:667-678.
3. Rinaldo CR Jr, Carney WP, Richter BS, Black PH, Hirsch MS. Mechanisms of immunosuppression in cytomegaloviral mononucleosis. *J Infect Dis*. 1980;141:488-495.
4. Drew WL, Mintz L, Miner RC, Sands M, Ketterer B. Prevalence of cytomegalovirus infection in homosexual men. *J Infect Dis*. 1981;143:188-192.
5. Lang DJ, Kummer JF. Cytomegalovirus in semen: observations in selected populations. *J Infect Dis*. 1975;132:472-473.

EPIDEMIOLOGIC NOTES AND REPORTS

Kaposi's Sarcoma and *Pneumocystis* Pneumonia Among Homosexual Men — New York City and California

July 3, 1981 · Vol. 30, No. 25 · pp. 305–308

During the past 30 months, Kaposi's sarcoma (KS), an uncommonly reported malignancy in the United States, has been diagnosed in 26 homosexual men (20 in New York City [NYC]; 6 in California). The 26 patients range in age from 26-51 years (mean 39 years). Eight of these patients died (7 in NYC, 1 in California)—all 8 within 24 months after KS was diagnosed. The diagnoses in all 26 cases were based on histopathological examination of skin lesions, lymph nodes, or tumor in other organs. Twenty-five of the 26 patients were white, 1 was black. Presenting complaints from 20 of these patients are shown in Table 1.

Skin or mucous membrane lesions, often dark blue to violaceous plaques or nodules, were present in most of the patients on their initial physician visit. However, these lesions were not always present and often were considered benign by the patient and his physician.

A review of the New York University Coordinated Cancer Registry for KS in men under age 50 revealed no cases from 1970-1979 at Bellevue Hospital and 3 cases in this age group at the New York University Hospital from 1961-1979.

Seven KS patients had serious infections diagnosed after their initial physician visit. Six patients had pneumonia (4 biopsy confirmed as due to *Pneumocystis carinii* [PC]), and one had necrotizing toxoplasmosis of the central nervous system. One of the patients with *Pneumocystis* pneumonia also experienced severe, recurrent, herpes simplex infection; extensive candidiasis; and cryptococcal meningitis. The results of tests for cytomegalovirus (CMV) infection were available for 12

Table 1. Presenting complaints in 20 patients with Kaposi's sarcoma

Presenting complaint	Number (percentage) of patients
Skin lesion(s) only	10 (50%)
Skin lesions plus lymphadenopathy	4 (20%)
Oral mucosal lesion only	1 (5%)
Inguinal adenopathy plus perirectal abscess	1 (5%)
Weight loss and fever	2 (10%)
Weight loss, fever, and pneumonia (one due to *Pneumocystis carinii*)	2 (10%)

patients. All 12 had serological evidence of past or present CMV infection. In 3 patients for whom culture results were available, CMV was isolated from blood, urine and/or lung of all 3. Past infections with amebiasis and hepatitis were commonly reported.

Since the previous report of 5 cases of *Pneumocystis* pneumonia in homosexual men from Los Angeles (1), 10 additional cases (4 in Los Angeles and 6 in the San Francisco Bay area) of biopsy-confirmed PC pneumonia have been identified in homosexual men in the state. Two of the 10 patients also have KS. This brings the total number of *Pneumocystis* cases among homosexual men in California to 15 since September 1979. Patients range in age from 25 to 46 years.

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Editorial Note: KS is a malignant neoplasm manifested primarily by multiple vascular nodules in the skin and other organs. The disease is multifocal, with a course ranging from indolent, with only skin manifestations, to fulminant, with extensive visceral involvement (2).

Accurate incidence and mortality rates for KS are not available for the United States, but the annual incidence has been estimated between 0.02-0.06 per 100,000; it affects primarily elderly males (3, 4). In a series of 92 patients treated between 1949 and 1975 at the Memorial Sloan-Kettering Cancer Institute in NYC, 76% were male, and the mean age was 63 years (range 23-90 years) at the time of diagnosis (5).

The disease in elderly men is usually manifested by skin lesions and a chronic clinical course (mean survival time is 8-13 years) (2). Two exceptions to this epidemiologic pattern have been noted previously. The first occurs in an endemic belt across equatorial Africa, where KS commonly affects children and young adults and accounts for up to 9% of all cancers (3). Secondly, the disease appears to have a higher incidence in renal transplant recipients (6–9) and in others receiving immunosuppressive therapy (10–12).

The occurrence of this number of KS cases during a 30-month period among young, homosexual men is considered highly unusual. No previous association between KS and sexual preference has been reported. The fulminant clinical course reported in many of these patients also differs from that classically described for elderly persons.

The histopathologic diagnosis of KS may be difficult for 2 reasons. Changes in some lesions may be interpreted as nonspecific, and other cutaneous and soft tissue sarcomas, such as angiosarcoma of the skin, may be confused with KS (13, 14).

That 10 new cases of *Pneumocystis* pneumonia have been identified in homosexual men suggests that the 5 previously reported cases were not an isolated phenomenon (1). In addition, CDC has a report of 4 homosexual men in NYC who developed severe, progressive, perianal herpes simplex infections and had evidence of cellular immunodeficiencies. Three died, 1 with systemic CMV infection. The fourth patient is currently undergoing therapy. It is not clear if or how the clustering of KS, pneumocystis, and other serious diseases in homosexual men is related. What is known is that the patients with *Pneumocystis* pneumonia described in the previous report showed evidence of impaired cellular immunity and previous or current CMV infection (1). Furthermore, serologic evidence of past CMV infection and active shedding of CMV have been shown to be much more common among homosexual men than heterosexual men attending a sexually transmitted disease clinic (15). A specific serologic association with CMV infection has been demonstrated among American and European patients with KS (16, 17) and herpes-type virus particles have been demonstrated in tissue culture cell lines from African cases of KS (18). It has been hypothesized that activation of oncogenic virus during periods of immunosuppression may result in the development of KS (19). Although immunosuppression often results in CMV infection, it is not yet clear whether CMV infection precedes or follows the above-mentioned disorders.

Although it is not certain that the increase in KS and PC pneumonia is restricted to homosexual men, the vast majority of recent cases have been reported from this group. Physicians should be alert for Kaposi's sarcoma, PC pneumonia, and other opportunistic infections associated with immunosuppression in homosexual men.

REFERENCES

1. Centers for Disease Control. *Pneumocystis* pneumonia—Los Angeles. *MMWR*. 1981;30:250.
2. Safai B, Good RA. Kaposi's sarcoma: a review and recent developments. *CA*. 1981;31:1-12.
3. Oettle AG. Geographical and racial differences in the frequency of Kaposi's sarcoma as evidence of environmental or genetic causes. *Acta Un Int Cancr*. 1962;18:330-363.
4. Rothman S. Remarks on sex, age, and racial distribution of Kaposi's sarcoma and on possible pathogenetic factors. *Acta Un Int Cancr*. 1962;18:326-329.
5. Safai B, Miké V, Giraldo G, Beth E, Good RA. Association of Kaposi's sarcoma with second primary malignancies: possible etiopathogenic implications. *Cancer*. 1980;45:1472-1479.
6. Harwood AR, Osoba D, Hofstader SL, et al. Kaposi's sarcoma in recipients of renal transplants. *Am J Med*. 1979;67:759-765.
7. Stribling J, Weitzner S, Smith GV. Kaposi sarcoma in renal allograft recipients. *Cancer*. 1978;42:442-446.
8. Myers BD, Kessler E, Levi J, Pick A, Rosenfeld JB, Tikvah P. Kaposi sarcoma in kidney transplant recipients. *Arch Intern Med*. 1974;133:307-311.
9. Penn I. Kaposi's sarcoma in organ transplant recipients: report of 20 cases. *Transplantation*. 1979;27:8-11.
10. Gange RW, Jones EW. Kaposi's sarcoma and immunosuppressive therapy: an appraisal. *Clin Exp Dermatol*. 1978;3:135-146.
11. Klepp O, Dahl O, Stenwig JT. Association of Kaposi's sarcoma and prior immunosuppressive therapy: a 5-year material of Kaposi's sarcoma in Norway. *Cancer*. 1978;42:2626-2630.
12. Hoshaw RA, Schwartz RA. Kaposi's sarcoma after immunosuppressive therapy with prednisone. *Arch Dermatol*. 1980;116:1280-1282.
13. Girard C, Johnson WC, Graham JH. Cutaneous angiosarcoma. *Cancer*. 1970;26:868-883.
14. Rosai J, Sumner HW, Kostianovsky M, Perez-Mesa C. Angiosarcoma of the skin. A clinicopathologic and fine structural study. *Hum Pathol*. 1976;7:83-109.
15. Drew WL, Mintz L, Miner RC, Sands M, Ketterer B. Prevalence of cytomegalovirus infection in homosexual men. *J Infect Dis*. 1981;143:188-192.
16. Giraldo G, Beth E, Kourilsky FM, et al. Antibody patterns to herpesvirus in Kaposi's sarcoma: serologic association of European Kaposi's sarcoma with cytomegalovirus. *Int J Cancer*. 1975;15:839-848.
17. Giraldo G, Beth E, Henle W, et al. Antibody patterns to herpesvirus in Kaposi's sarcoma. II. Serological association of American Kaposi's sarcoma with cytomegalovirus. *Int J Cancer*. 1978;22:126-131.
18. Giraldo G, Beth E, Haguenau F. Herpes-type virus particles in tissue culture of Kaposi's sarcoma from different geographic regions. *J Natl Cancer Inst*. 1972;49:1509-1526.
19. Kapadia SB, Krause JR. Kaposi's sarcoma after long-term alkylating agent therapy for multiple myeloma. *South Med J*. 1977;70:1011-1013.

FROM THE ISSUE OF JULY 3, 1981

Notifiable Diseases — United States

25th week ending June 27, 1981 · Vol. 30, No. 25 · p. 308

Each weekly issue carried the counts of notifiable diseases that the states reported to CDC. The summary table of the July 3 issue is reprinted as it was printed (Table 1).

Table 1. Summary — cases of specified notifiable diseases, United States

Disease	25th week ending		Median 1976–1980	Cumulative, first 25 weeks		
	June 27, 1981	June 21, 1980		June 27, 1981	June 21, 1980	Median 1976–1980
Aseptic meningitis	133	107	107	1,859	1,619	1,126
Brucellosis	3	5	6	73	82	82
Chickenpox	3,801	4,577	4,353	157,690	144,465	144,465
Diphtheria	–	–	–	3	2	35
Encephalitis: Primary (arthropod-borne & unspec.)	20	8	18	360	282	287
Encephalitis: Post-infectious	2	4	4	44	98	102
Hepatitis, Viral: Type B	404	364	317	9,425	7,977	7,262
Hepatitis, Viral: Type A	430	579	595	11,981	12,945	14,023
Hepatitis, Viral: Type unspecified	185	214	178	5,367	5,296	4,253
Malaria	43	58	19	634	853	265
Measles (rubeola)	85	463	924	2,292	11,224	20,679
Meningococcal infections: Total	48	50	40	2,021	1,533	1,357
Meningococcal infections: Civilian	46	49	39	2,009	1,522	1,299
Meningococcal infections: Military	2	1	1	12	11	21
Mumps	54	158	422	2,665	6,382	11,698
Pertussis	23	26	26	482	524	524
Rubella (German measles)	30	82	362	1,443	2,756	9,753
Tetanus	4	4	1	27	29	29
Tuberculosis	505	613	689	12,863	12,672	13,778
Tularemia	8	7	4	87	70	62
Typhoid fever	7	10	8	224	171	171
Typhus fever, tick-borne (Rky. Mt. spotted)	57	39	48	436	318	299
Venereal diseases:						
Gonorrhea: Civilian	18,174	20,192	20,192	463,413	452,245	452,245
Gonorrhea: Military	579	368	495	13,731	12,756	12,844
Syphilis, primary & secondary: Civilian	610	488	452	14,248	12,387	11,575
Syphilis, primary & secondary: Military	6	5	5	178	151	146
Rabies in animals	162	136	73	3,479	3,250	1,509

Cumulative totals include revised and delayed reports through previous weeks.

Surveillance Notes, Reprint No. 1. Re-set from the printed issues of June 5 and July 3, 1981 (Morbidity and Mortality Weekly Report, vol. 30, nos. 21 and 25, Centers for Disease Control, Atlanta), which are in the public domain. The text and the tables keep the wording and figures of 1981, including the terminology of the time. Figure 1 and the case table on page 2 were added by the editors. Set in IBM Plex Serif, Libre Franklin and IBM Plex Sans Condensed (SIL Open Font License).